

AI Data Platform / Lakehouse

Bronze/Silver/Gold + Feature Store + MLflow-ready forecasting + Governance

Project Question

Can a modern healthcare/public-sector lakehouse improve scalable AI analytics and forecasting workflows?

Overview Data Quality Forecasting **Architecture**

Lakehouse Architecture

Raw public-sector / healthcare-style events

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Streaming simulator

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Bronze Layer

Raw immutable events

↓

Silver Layer

Cleaned, standardized, de-identified records

↓

Gold Layer

Certified analytics datasets

↓

```
graph TD; A[Feature Store] --> B[Reusable ML features]; B --> C[MLflow Forecasting Model]; C --> D[Dashboard + Governance Report];
```

Feature Store
Reusable ML features
↓
MLflow Forecasting Model
↓
Dashboard + Governance Report

Layers

Bronze

Raw immutable data saved as parquet.

Silver

Cleaned data with standardized schema, missing-value handling, and de-identification-safe fields.

Gold

Certified business-ready tables for analytics and reporting.

Feature Store

ML-ready features used by forecasting models.

Governance Layer

Includes schema checks, quality checks, lineage notes, and privacy guidance.

Governance and Privacy

This package uses synthetic healthcare/public-sector style data.

Public GitHub Safety Rules

Do not commit:

- raw PII
- names
- birth dates
- addresses
- student or patient identifiers
- secure warehouse paths
- credentials
- internal production extracts

Recommended production governance

- role-based access control
- de-identification
- schema validation
- table lineage
- certified dataset ownership
- audit logs

- model monitoring
- MLflow model registry
- data quality SLA checks